

Peer relationships and adolescent internet addiction: Variable-centered and person-centered approaches

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ABSTRACT

This research aimed to examine the relationship between peer relationships (i.e., popularity, peer rejection, and loneliness) and internet addiction among Chinese adolescents, as well as explore the underlying mechanisms. Cross-sectional data from 1,388 Chinese adolescents, including 46.4 % girls, with a mean age of 14.53 years (± 1.98 years), was analyzed. Latent profiles analysis identified three peer relationship profiles: the popular peer status profile (61.2 %), the moderate peer status profile (32.2 %), and the unpopular peer status profile (6.6 %). The mediating roles of self-annoyance and self-control were found in the associations between peer relationships and internet addiction using both variable and person-centered approaches. These findings hold significant implications for tailoring intervention programs that target the prevention of internet addiction among adolescents with poor peer relationships.

1. Introduction

The advent of the Internet era has brought numerous opportunities and benefits to people's lives, such as online shopping, education, leisure activities, and telemedicine (Ostovar et al., 2021). However, with increased access to the Internet, adolescents are at a higher risk of developing internet addiction (Mo et al., 2020). Internet addiction is typically characterized by the inability to manage or control one's internet usage (Cheng & Li, 2014). Its prevalence among adolescents has been reported to be high in various regions, such as the United States (Cheng & Li, 2014), Europe (Kapus et al., 2021), and Asian countries (Xin et al., 2018). Symptoms of internet addiction often include uncontrolled impulses, tolerance, and withdrawal (Chang et al., 2015; Ostovar et al., 2021; Stodt et al., 2016). Additionally, excessive use of social networks like Twitter, Facebook, and WeChat has been linked to increased real-life isolation (Guedes et al., 2016; Müller et al., 2016), which may further contribute to excessive internet use (Müller et al., 2016).

This has prompted a growing interest in studying the factors that lead to internet addiction (Burnay et al., 2015; Malak et al., 2017). Peer context has a stronger impact on adolescents than any other period in their lives, often exceeding the influence of their family environment (e.g., Fomby & Sennott, 2013). Consequently, considerable attention has

been devoted to understanding the relationship between peer relationships and adolescents' internet addiction (e.g., Li et al., 2023; Savcı & Aysan, 2017; Soh et al., 2018). A consensus has emerged that negative peer relationships exacerbate the risk of internet addiction (e.g., Aziz et al., 2022; Zhou et al., 2017). However, traditional variable-centered approach presents challenges when examining the multidimensional nature of peer relationships (Engels et al., 2019) in relation to internet addiction. This approach may not adequately capture the unique combinations of peer relationship dimensions, leading to a potential oversight of individual differences, hampering the ability to identify specific subgroups within the population that may require targeted and effective interventions. To address these issues, the present study utilized both variable-centered and person-centered approaches in a single study, grounded in the compensatory internet use theory (Kardefelt-Winther, 2014a), to examine the connections between peer relationships and internet addiction among adolescents, along with exploring the underlying mechanisms, which may be helpful in obtaining a more comprehensive delineation of this phenomenon.

1.1. Peer relationships and internet addiction: The compensatory internet use theory

The compensatory internet use theory, proposed by Kardefelt-

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Winther (2014a), suggests that internet addiction can function as a coping mechanism employed to address adverse life circumstances or alleviate negative emotional states. This means that individuals who lack sufficient social stimulation may be more motivated to engage in online activities (e.g., video games or social networking sites) (Melodia et al., 2022). Therefore, based on this theory, adolescents may develop internet addiction as a means to compensate for their poor peer relationships.

During adolescence, there is a shift in focus from the family to the peer group, and adolescents interact in increasingly diverse social environments. The social world and peer interactions become increasingly important during this developmental stage (Orben et al., 2020). Poor peer relationships, such as feeling rejected by peers, can hinder the satisfaction of the basic need for relatedness (Ryan & Deci, 2017). This can increase the risk of mental health issues (Platt et al., 2013), impair the development of social brains (Orben et al., 2020), and hinder the acquisition of social skills such as cooperation and responsiveness (Rubin et al., 2005). The internet can help meet social needs for individuals who struggle to build relationships in traditional ways. On the one hand, the Internet offers a platform that enhances social connectedness, provides a sense of belonging, and modulates negative moods associated with loneliness (Liu & Kuo, 2007; Morahan-Martin & Schumacher, 2003); On the other hand, the Internet offers a level of anonymity, providing opportunities for self-expression through relative freedom from possible constraints (Keipi et al., 2015). These might enhance online friendship formation and increase the risk of developing internet addiction (Amichai-Hamburger et al., 2013; Smahel et al., 2012). Consistent with this notion, research indicates that adolescents who experience difficulties in their peer relationships, such as social rejection and loneliness, are more likely to develop internet addiction as they try to compensate for the absence of fulfilling offline social connections (e.g., Morahan-Martin & Schumacher, 2003; Wang et al., 2017).

Therefore, based on the compensatory internet use theory and existing literature, we put forward the following hypothesis:

H1: Poor peer relationships will be positively associated with internet addiction.

1.2. Mechanisms between peer relationships and internet addiction

1.2.1. Self-annoyance

Self-annoyance refers to an individual's negative self-evaluation accompanied by negative emotions, often arising from feelings of personal incompetence, perceived failures, or unmet expectations (Chen et al., 2019; Zou et al., 2012). Drawing upon the social identity theory (Tajfel, 1978), it is understood that individuals define their self-concept based on social categories or group memberships. During adolescence, when individuals face rejection and exclusion from peers, their understanding of their own identity and their place within social contexts can be challenged (Buhs, 2005; Tanti et al., 2011). Consequently, individuals may experience self-annoyance, engaging in self-criticism or self-betittlement due to their perceived failure to meet social expectations or gain acceptance from their peers. In search of a more positive self-concept, individuals may turn to the Internet, which might serve as a compensatory platform where individuals can form new connections, seek social support, and construct a self-identity that may be lacking in their offline peer interactions (Gao et al., 2017; Košir et al., 2016; Wang & Wang, 2013). In line with this idea, studies have found that self-esteem plays a role in explaining how peer relationships are connected to internet or smartphone addiction among adolescents (Park et al., 2014; Wang et al., 2017).

Based on established findings, we hypothesize that:

H2: The association between peer relationships and internet addiction can be mediated by self-annoyance.

1.2.2. Self-control

Self-control refers to the capacity to resist impulses and temptations and to actively and consciously modify one's behavior (Hagger, 2010). The strength model of self-control (Maranges & Baumeister, 2016) suggest that self-control is a limited resource that can become depleted with use. This means that using self-control in one task or situation can reduce the ability to exercise self-control later, resulting in decreased performance in subsequent acts of self-control (Hagger et al., 2010).

Interpersonal interactions, such as peer relationships, often require self-control (Muraven et al., 2019). For adolescents facing poor peer relationships, such as conflicts or rejection, constant exertion of self-control can deplete their self-regulatory capacity. This depletion of self-control can then make it more challenging for them to exercise self-control in other areas of their lives, potentially leading to engagement in risky behaviors like crime or gambling (Maranges & Baumeister, 2016). In the context of the compensatory internet use theory, it's suggested that internet addiction might act as a way for individuals who find it challenging to control themselves in real-life situations to compensate for their lack of self-control. The online world, with its accessibility, anonymity, and immediate gratification, may provide an outlet for individuals who find it difficult to exert self-control in offline domains. This notion finds support in research indicating that adolescents with low self-control in offline contexts are more prone to developing internet addiction (Li et al., 2014; Özdemir et al., 2014). Therefore, based on the aforementioned theories and findings, we suggest that:

H3: The association between peer relationships and internet addiction can be mediated by self-control.

Furthermore, individuals experiencing self-annoyance often engage in self-criticism and negative thoughts, and focus on perceived failures or shortcomings (Chen et al., 2019). According to the strength model of self-control (Maranges & Baumeister, 2016), this negative self-evaluation can deplete the cognitive resources necessary for effective self-control, while positive self-evaluations are positively associated with self-control. Consistent with this notion, Tangney et al. (2004) proposed that "people with high self-control apparently accept themselves as valuable, worthy individuals and are relatively well able to sustain this favorable view of self across time and circumstances" (p. 299), which is supported by empirical evidence (Roberts et al., 2015; Xiang et al., 2022). Moreover, indirect evidence suggests that self-affirmation (a contrasting process to self-annoyance) and positive mood induction could be effective in mitigating these depleting effects (Schmeichel & Vohs, 2009; Tice et al., 2007). Therefore, we propose the following hypothesis based on the interplay between peer relationships, self-annoyance, self-control, and previous findings:

H4: Self-annoyance and self-control will act as sequential mediators in the association between peer relationships and adolescents' internet addiction.

1.3. Combining the variable-centered and person-centered approaches

To date, most research on the relationship between peer relationships and internet addiction has primarily adopted a variable-centered approach. This approach focuses on how different dimensions of peer relationships are uniquely and independently related to specific outcomes. However, it is essential to recognize that peer relationships are multi-dimensional, and studies that rely on a traditional variable-centered approach often overlook the possibility that adolescents may experience distinct patterns of peer relationships. For instance, loneliness, which refers to the subjective experience of deficiencies in social relationships in terms of quantity and/or quality, is conceptually different from physical isolation (Mund et al., 2022). Research has found an intriguing phenomenon where certain popular children report remarkably high levels of loneliness, while some rejected children report

surprisingly low levels of loneliness (Lau et al., 1999).

In contrast, person-centered approach is based on the underlying principle that, instead of quantifying the role of specific variables in a study, a population is organized into a finite number of mutually exclusive and exhaustive subgroups, each consisting of similar individuals (Lanza & Cooper, 2016; Lanza & Rhoades, 2013). Therefore, in the present study, the person-centered approach would extend the findings from the variable-centered approach by revealing how adolescents could be classified into homogenous groups with similar peer relationship profiles and how these groups vary in terms of internet addiction.

Prior research has provided valuable insights into the importance of adopting a person-centered approach when studying the influence of peer relationships on outcomes (Chen et al., 2021; Engels et al., 2019; Favre et al., 2022; Lan et al., 2023). For example, Lan et al. (2023) identified three profiles in adolescence: “social competent” (16.3 %), “average” (69.1 %), and “isolated” (14.6 %). Similarly, other studies have reported four peer relationship profiles (e.g., Chen et al., 2021; Favre et al., 2022). However, there is a scarcity of studies that have utilized a person-centered approach to examine the association between peer relationship profiles and internet addiction. Variable-centered and person-centered approaches are mutually complementary (Masyn, 2013). By combining these two approaches, a more comprehensive understanding of the relationship between peer relationships and internet addiction, as well as the underlying mechanisms, can be unveiled.

H5: There will be at least three distinct profiles of peer relationships that would be associated with internet addiction.

H6: Self-annoyance and self-control will mediate the association between peer relationship profiles and adolescents' internet addiction.

1.4. The present study

China has a significant population of adolescent Internet users, and internet addiction is prevalent among this demographic (Yu et al., 2019). In an effort to enhance the existing body of knowledge, the present study sought to examine the following aspects by adopting both variable-centered and person-centered approaches: (1) examine the direct effects of peer relationships on adolescents' internet addiction; (2) explore the potential mediating effects of self-annoyance and self-control in the associations between peer relationships and internet addiction; (3) examine the potential association between various patterns of peer relationships and internet addiction among Chinese adolescents using a person-centered approach; and (4) investigate the mediating roles of self-annoyance and self-control in the relationship between peer relationship profiles and internet addiction.

2. Method

2.1. Participants

This study is part of a larger research project concerning adolescents' internet use (see Jin et al., 2019). A sample of 1,388 middle and high school students (46.4 % girls) with an average of 14.53 years (± 1.98 years) from three public schools in Beijing, China, participated in this study. The majority of students (89.8 %) identified as Han descent. In regards to parents' highest level of educational attainment, about half of the adolescents reported their mothers (45.9 %) and fathers (52.4 %) had at least a college degree, while over 30 % of adolescents indicated their mothers (37.0 %) and fathers (34.4 %) had a high school diploma. Furthermore, the median monthly household income ranged between CNY 8,000 and CNY 9,999.

2.2. Measures

2.2.1. Peer relationships

Peer relationships were assessed using Zhang's (2008) revised Chinese version of the Loneliness and Dissatisfaction Questionnaire (LDQ) developed by Asher et al. (1984). The LDQ is a self-report scale with 16 items to measure children's satisfaction with their peer relationships. It consists of three dimensions: popularity, peer rejection, and loneliness. Participants rated their responses on a 4-point scale ranging from 1 (Strongly disagree) to 4 (Strongly agree). After reverse-scoring certain items, the mean score was calculated for each dimension, with higher scores indicating a higher level of that dimension. The Cronbach's alpha ranged from 0.80 to 0.84 for the three subscales.

2.2.2. Self-annoyance

The self-annoyance was measured using Yu's (2009) revised Chinese adolescent self-annoyance subscale, based on the Chinese Adolescent Social Adjustment Scale developed by Zhou et al. (2008). The scale consists of 8 items, and participants rated their responses on a 5-point scale ranging from 1 (totally disagree) to 5 (totally agree). The mean score was calculated for each item, with higher scores indicating a higher level of self-annoyance. The Cronbach's alpha for the scale in the current study was 0.93.

2.2.3. Self-control

Self-control was measured using Tan and Guo (2008) revised Chinese version of the Self-Control Questionnaire (SCQ) developed by Tangney (2004). The scale contains 19 items that assess various aspects of self-control, including impulse control, healthy habits, resisting temptation, concentration on work, and temperance in entertainment. Participants rated their agreement on a 5-point scale ranging from 1 (totally disagree) to 5 (totally agree). After reverse-scoring certain items, we computed the mean score for all 19 items, with higher scores reflecting greater levels of self-control. The Cronbach's alpha for the scale was 0.89, and for the subscales, it ranged from 0.59 to 0.83.

2.2.4. Internet addiction

Internet addiction was measured using the 10-item Diagnostic Questionnaire developed by Young (1998). Each item was scored as 0 (no) or 1 (yes). The sum of the ten items was taken, with higher scores indicating higher levels of internet addiction. The Cronbach's alpha for the questionnaire in the current study was 0.80.

2.2.5. Control variables

Gender (0 = boy; 1 = girl) and socioeconomic status (SES) were included as control variables in the mediation analyses. SES was assessed by constructing a composite variable that considered the mother's educational level, the father's educational level, and the monthly household income. Each component of SES was standardized, and then the standardized scores were subsequently averaged to create a final composite score (Matthews et al., 2010).

2.3. Procedure

Adolescents were provided with comprehensive information regarding the purpose of the study and were invited to participate on a voluntary basis. Written consent from both parents and assent from the students were required for participation. The surveys were administered by trained teachers, and the participating adolescents completed them individually during a 45-minute class period in their respective classrooms. The study followed the ethical principles outlined by the Code of Ethics Review Committee for Human Psychology Research at Beijing Forestry University.

2.4. Data analysis

IBM SPSS Statistics (Version 25) and Mplus 8.3 were utilized for data processing and analysis. In our *variable-centered analyses*, we first conducted preliminary analyses (mean and standard deviations, and intercorrelations). Second, the direct effects of peer relationship indicators (i.e., popularity, peer rejection, and loneliness) on internet addiction were examined while controlling for gender and SES through stepwise multiple regressions. Third, we applied a structural equation model (SEM) to investigate the mediating roles of self-annoyance and self-control in the associations between peer relationships and internet addiction, while controlling for gender and SES. A series of fit indices were evaluated to estimate the model fit: the chi-square goodness-of-fit test (χ^2), Comparative Fit Index (CFI), the Tucker-Lewis Index (TLI), and the root mean square error of approximation (RMSEA). We considered the following indices as evidence of a good model: TLI > 0.90, CFI > 0.90, and RMSEA < 0.08. To test the mediation effects, we used the bootstrapping method with 1,000 resamples within a 95 % confidence interval.

In our *person-centered analyses*, we first utilized latent profile analysis (LPA) to determine the optimal number of latent profiles of peer relationships based on the three indicators (i.e., popularity, peer rejection, and loneliness). The selection of the optimal model fit was based on lower Akaike information criterion (AIC) values and Bayesian information criterion (BIC) values, higher entropy, and a significant Lo-Mendell-Rubin adjusted likelihood ratio test (LMR-LRT) (Akaike, 1974; Schwarz, 1978). Additionally, the profile size should be meaningful (Nylund et al., 2007); if a specific profile constituted less than 5.0 % of the total sample size, the profile should be rejected. After determining the number of profiles, multivariate analysis of variance (MANOVA) and post hoc tests (Bonferroni correction) were conducted to determine whether latent peer relationship profiles differed significantly in popularity, peer rejection, and loneliness. Furthermore, a one-way analysis of variance (one-way ANOVA) was performed to investigate mean differences in internet addiction among the peer relationship profiles. Additionally, SEM was applied to investigate how self-annoyance and self-control mediate the connections between latent peer relationship profiles and internet addiction, while controlling for gender and SES. Indirect effects were also examined using a bootstrapping method with 1,000 resamples to compute the indirect effects and obtain 95 % confidence intervals. Finally, since latent peer relationship profiles encapsulated the interactions of three peer relationship indicators, we included both these indicators and the latent peer relationship profiles simultaneously in a mediation model. This allowed us to test whether these profiles retained their significance even after controlling for the main effects of the indicators.

Table 1
Descriptive statistics and intercorrelations among study variables.

	1	2	3	4	5	6	7	8
1. Gender (0 = boy)	1							
2. SES	−0.03	1						
3. Popularity	−0.01	0.11***	1					
4. Loneliness	0.01	−0.14***	−0.72***	1				
5. Peer rejection	−0.003	−0.09**	−0.71***	0.81***	1			
6. Self-control	−0.04	0.11***	0.34***	−0.41***	−0.38***	1		
7. Self-annoyance	0.05	−0.13***	−0.44***	0.60***	0.54***	−0.52***	1	
8. Internet addiction	−0.002	−0.07**	−0.21***	0.28***	0.27***	−0.61***	0.40***	1
M	0.47	0.0002	3.17	1.68	1.51	3.57	2.05	2.46
SD	0.50	0.82	0.64	0.66	0.61	0.71	1.01	2.47

Note. ** $p < 0.01$, *** $p < 0.001$.

3. Results

3.1. Variable-centered approach

3.1.1. Preliminary analyses

The means, standard deviations, and intercorrelations for the study variables were calculated and summarized in Table 1. Popularity exhibited negative associations with both peer rejection and loneliness. Furthermore, popularity showed negative correlations with self-annoyance and internet addiction, but displayed a positive relationship with self-control. In contrast, the relationships between peer rejection, loneliness, and these variables were opposite to those observed with popularity.

3.1.2. Direct effects of peer relationships on internet addiction

The direct effects of peer relationship indicators on adolescents' internet addiction were examined while controlling for gender and SES (see Table 2). In Model 2, popularity exhibited a negative association with internet addiction. However, with the inclusion of peer rejection and loneliness in Model 3 and Model 4, respectively, these factors demonstrated a positive relationship with internet addiction. Notably, in this context, popularity ceased to maintain a significant association with internet addiction.

3.1.3. Indirect effects of peer relationships on internet addiction

As shown in Fig. 1, after controlling for gender and SES, the mediation model exhibited a good fit to the data: $\chi^2 = 14.61$, $df = 4$, TLI = 0.97, CFI = 0.99, RMSEA = 0.04. To test the significance of the mediation model, we conducted a bootstrapping procedure with 1,000 resamples. The following mediation paths were identified: (1) Self-annoyance fully mediated the effect of loneliness on internet addiction (effect size = 0.06, 95 % CI [0.03, 0.10]); (2) Self-annoyance also fully mediated the effect of peer rejection on adolescent's internet addiction (effect size = 0.02, 95 % CI [0.01, 0.04]); (3) The chain mediation effect of self-annoyance and self-control on the link between loneliness and internet addiction was significant (effect size = 0.11, 95 % CI [0.09, 0.14]); (4) The chain mediation effect of self-annoyance and self-control on the link between peer rejection and internet addiction was also significant (effect size = 0.04, 95 % CI [0.02, 0.07]). However, regarding popularity, we did not find a significant mediation effect on internet addiction.

3.2. Person-centered approach

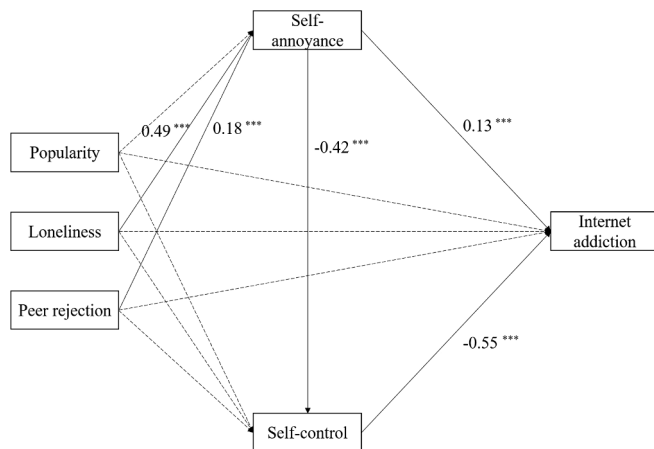
3.2.1. Latent profiles of peer relationships and internet addiction

To facilitate a clear interpretation of indicator values in relation to the sample means, the scores of peer relationship indicators (i.e., popularity, peer rejection, and loneliness) were z-score transformed before conducting the LPA based on these indicators. Details of fit indices and profile information for all profile results are presented in Table 3. The results indicated a continuous decrease in AIC and BIC

Table 2

The prediction of peer relationship indicators on internet addiction.

	Model 1		Model 2		Model 3		Model 4	
	β	t	β	t	β	t	β	t
Gender (0 = boy)	–0.01	–0.31	–0.01	–0.25	–0.01	–0.30	–0.01	–0.25
SES	–0.08	–2.86**	–0.05	–1.60	–0.04	–1.46	–0.04	–1.60
R^2	0.01							
Popularity			–0.20	–7.50***	–0.01	–0.28	0.03	0.71
R^2			0.05					
Loneliness					0.27	7.21***	0.18	3.91***
R^2					0.08			
Peer rejection							0.14	3.04**
R^2							0.09	

Note. ** $p < 0.01$, *** $p < 0.001$.**Fig. 1.** Standardized regression coefficients in the mediation SEM model through variable-centered approach. Note. *** $p < 0.001$. Control variables are SES and gender. A dashed line indicates a nonsignificant path.**Table 3**

Model fit indices of latent profile analysis.

Profile	AIC	BIC	Entropy	LMR-LRT p -value	Number of subjects per class
2	9834.00	9886.35	0.89	< 0.001	442/946
3	8941.99	9015.28	0.92	< 0.001	92/447/849
4	8655.17	8749.41	0.91	0.003	44/158/388/798
5	8457.16	8572.35	0.92	0.03	35/100/255/294/704

values from the 2- to the 5-profile model. However, the improvement in AIC and BIC values became smaller after the 3-profile model. The LMR-LRT p -values were significant for all models, indicating a better fit with increasing profile numbers. In addition, it's noteworthy that the 3-profile and 5-profile models displayed the highest entropy value. Nevertheless, some of the 4- and 5-profile models had profiles that accounted for less than 5.0 % of the total sample size, suggesting a lack of meaningfulness. Considering these factors, the 3-profile model was selected as the most appropriate.

Fig. 2 provides a visual representation of the distribution of the three latent profiles. Profile 1, referred to as the *moderate peer status* ($n = 447$, 32.2 %), represents adolescents with moderate levels of popularity, peer rejection, and loneliness. Profile 2, referred to as the *unpopular peer status* ($n = 92$, 6.6 %), represents adolescents who scored high on both loneliness and peer rejection, and low on popularity. Profile 3, referred to as the *popular peer status* ($n = 849$, 61.2 %), reflects adolescents with low levels of loneliness and peer rejection, and high levels of popularity. Significant differences were observed among these profiles in terms of popularity, peer rejection, and loneliness, as shown in Table 4.

In addition, Table 4 demonstrates that adolescents in the unpopular peer status profile reported the highest levels of internet addiction compared to the other two profiles. Additionally, significant differences were found between adolescents within the moderate peer status profile and those in the popular peer status profile.

3.2.2. Indirect effects of latent profiles of peer relationship on internet addiction

We conducted a mediation SEM to investigate how self-annoyance and self-control mediate the connections between latent peer relationship profiles and internet addiction, while also considering gender and SES as control variables. Given that the mediation analysis included a categorical predictor, the results represent group comparisons with the popular peer status profile, which was chosen as the reference group because it exhibited the lowest levels of internet addiction.

As illustrated in Fig. 3, after controlling for gender and SES, the mediation model showed a good fit to the data: $\chi^2 = 16.87$, $df = 4$, TLI = 0.97, CFI = 0.99, RMSEA = 0.05. We employed a bootstrapping procedure with 1,000 resamples to assess the statistical significance of the mediation model. Compared to adolescents in the popular status profile, the following mediation paths were found: (1) self-annoyance fully mediated the effect of the moderate peer status profile (effect size = 0.06, 95 % CI [0.03, 0.08]) and unpopular peer status profile (effect size = 0.06, 95 % CI [0.03, 0.09]) on adolescents' internet addiction; (2) self-control fully mediated the effect of the moderate peer status profile (effect size = 0.08, 95 % CI [0.04, 0.11]) and unpopular peer status profile (effect size = 0.04, 95 % CI [0.004, 0.08]) on adolescents' internet addiction; (3) the chain mediation effect of self-annoyance and self-control on the link between the moderate peer status profile (effect size = 0.11, 95 % CI [0.09, 0.13]) and unpopular peer status profile (effect size = 0.11, 95 % CI [0.08, 0.13]) and adolescents' internet addiction was also significant.

Finally, we simultaneously incorporated the three indicators (i.e., popularity, peer rejection, and loneliness) and latent peer relationship profiles (dummy coded) into the mediation model (see Fig. 4). After controlling for gender and SES, the mediation model also showed a good fit to the data: $\chi^2 = 12.49$, $df = 4$, TLI = 0.97, CFI = 0.99, RMSEA = 0.04. However, the bootstrapping mediation results were not significant for moderate peer status profile and unpopular peer status profile because the CIs included zero. Therefore, they are not reported. As the findings in variable-centered analyses, only the following mediation paths were identified: (1) Self-annoyance fully mediated the effect of loneliness on internet addiction (effect size = 0.06, 95 % CI [0.03, 0.09]); (2) Self-annoyance also fully mediated the effect of peer rejection on adolescent's internet addiction (effect size = 0.02, 95 % CI [0.01, 0.05]); (3) The chain mediation effect of self-annoyance and self-control on the link between loneliness and internet addiction was significant (effect size = 0.11, 95 % CI [0.08, 0.14]); (4) The chain mediation effect of self-annoyance and self-control on the link between peer rejection and internet addiction was also significant (effect size = 0.04, 95 % CI [0.01, 0.07]).

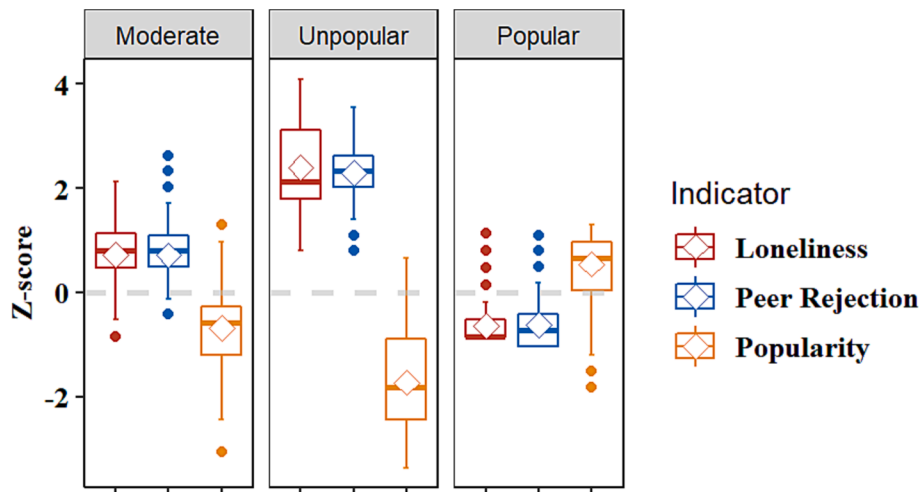


Fig. 2. Profiles of peer relationships. Note. Each plot in Fig. 2 depicts the distribution of data using a boxplot. The box represents the interquartile range, spanning from the 25th to the 75th percentile. The horizontal line within the box represents the median, and the diamond represents the mean. Whiskers extend to the smallest observations within 1.5 times the interquartile range below the 25th percentile, and any observations outside this range are represented by dots.

Table 4
Mean differences in study variables among three peer relationship profiles.

	Moderate Peer Status <i>M (SD)</i>	Unpopular Peer Status <i>M (SD)</i>	Popular Peer Status <i>M (SD)</i>	<i>F</i>	η_p^2
Popularity	-0.68 (0.72)	-1.73 (1.12)	0.55 (0.61)	774.05	0.53
Peer rejection	0.71 (0.53)	2.29 (0.62)	-0.62 (0.47)	2033.18	0.75
Loneliness	0.71 (0.54)	2.40 (0.86)	-0.64 (0.33)	2646.91	0.79
Internet addiction	3.15 (2.50)	3.84 (2.97)	1.95 (2.24)	53.86	0.07

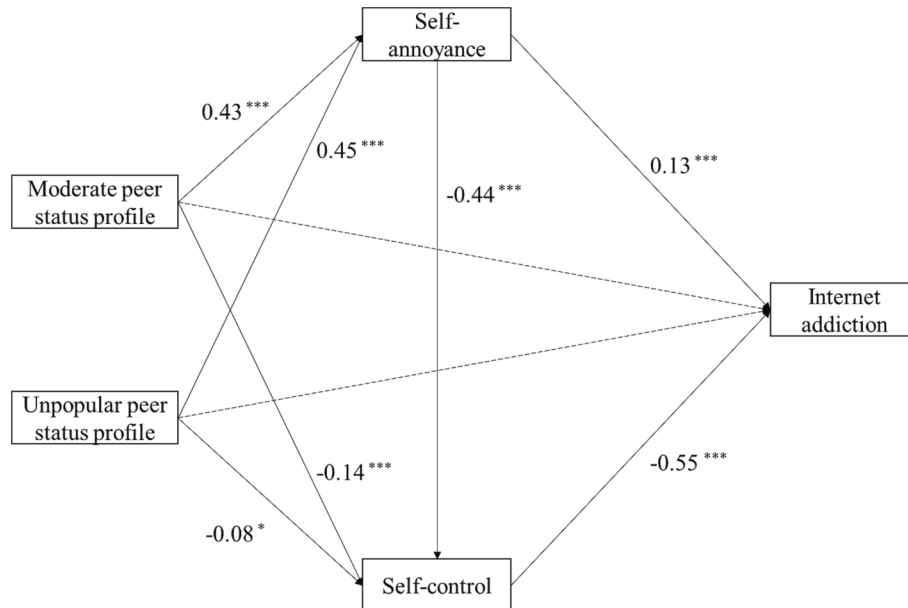


Fig. 3. Standardized regression coefficients in the mediation SEM model through person-centered approach. Note. * $p < 0.05$, *** $p < 0.001$. The reference group is the popular peer status profile. Control variables are SES and gender. A dashed line indicates a nonsignificant path.

4. Discussion

Based on the compensatory internet use theory (Kardefelt-Winther, 2014a), the present study examined the associations between peer relationship indicators (i.e., popularity, loneliness, and peer rejection) and internet addiction, as well as the mediating roles of self-annoyance and self-control using both variable-centered and person-centered

approach. The findings of this study offer valuable insights into the developmental importance of peer relationships and their connections to internet addiction in adolescents.

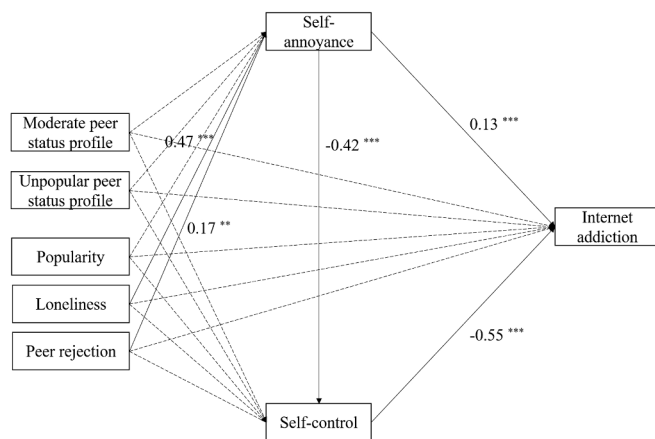


Fig. 4. Standardized regression coefficients in the mediation SEM model for integrating variable- and person-centered approaches. Note. ** $p < 0.01$, *** $p < 0.001$. The reference group is the popular peer status profile. Control variables are SES and gender. A dashed line indicates a nonsignificant path.

4.1. Synthesis of major findings

4.1.1. Variable-centered approach

Using a variable-centered approach, we quantified the unique and independent effects of peer relationship indicators while controlling for potential confounds. Supporting hypothesis H1, peer rejection and loneliness, two maladaptive peer relationship indicators, positively predicted adolescents' internet addiction. These results align with previous studies (e.g., Morahan-Martin & Schumacher, 2003; Wang et al., 2017) and are consistent with the compensatory internet use theory (Kardefelt-Winther, 2014a). This theory suggests that the Internet, by enhancing social connectedness and providing anonymity for self-expression (Liu & Kuo, 2007; Keipi et al., 2015), can help fulfill the social needs of adolescents with poor peer relationships, such as seeking acceptance and a sense of belonging, leading to excessive Internet use.

Notably, when popularity was examined in isolation, it demonstrated a protective role against internet addiction. However, this protective effect disappeared when we introduced peer rejection and loneliness into the analysis. To explain this shift, we can draw a parallel to the wooden barrel theory. Just as a barrel's capacity is determined by its shortest wooden stave, a protective factor's effectiveness may be gauged by its weakest link. Initially, when popularity was studied in isolation, it appeared to act as a safeguard against internet addiction, akin to a sturdy stave in the barrel. However, when we introduced peer rejection and loneliness into the analysis, they functioned as the shorter, weaker staves in our metaphorical barrel. Consequently, the overall protective effect of popularity waned.

Furthermore, this study developed and tested the underlying mechanisms within the compensatory framework. The results demonstrated that self-annoyance, rather than self-control, acted as an independent mediator, connecting both peer rejection and loneliness to adolescents' internet addiction. Importantly, a chain mediation model demonstrated that self-annoyance and self-control sequentially mediated the association between peer rejection and loneliness and adolescents' internet addiction. Thus, the findings supported hypotheses H2 and H4, while hypothesis H3 was not supported. Several reasons could explain the lack of a significant individual mediating effect of self-control. On the one hand, based on the compensatory internet use theory (Kardefelt-Winther, 2014a) and the strength model of self-control (Maranges & Baumeister, 2016), it is worth noting that when adolescents confront peer rejection and loneliness *in isolation*, they might not fully capture the effects of these challenges on self-control resources; On the other hand, during adolescence, social identity threat may take precedence. Individuals experiencing poor peer relationships may grapple with the

challenge of forming a sense of self within their peer groups (Tanti et al., 2011), thus rendering self-annoyance more relevant as a mediator in the relationship between peer rejection and loneliness and internet addiction, as opposed to self-control.

4.1.2. Person-centered approach

Supporting hypothesis H5, we identified three distinct peer relationship profiles (i.e., unpopular peer status, moderate peer status, and popular peer status) and found that adolescents classified in the unpopular peer status profile were at the highest risk of developing internet addiction, followed by those in the moderate peer status profile. These findings through a person-centered approach also align with the compensatory internet use theory (Kardefelt-Winther, 2014a). Furthermore, the degree of poor peer relationships appears to influence the intensity of the need to cope, with stronger needs leading to more online time (Kardefelt-Winther, 2014b).

In terms of underlying mechanisms, what stands out is our observation of the independent mediating effect of self-control within both the moderate peer status profile and the unpopular peer status profile, a phenomenon that did not emerge in the variable-centered approach. The distinction could be attributed to the person-centered approach's ability to capture the complexity and interactions among variables (Lanza & Cooper, 2016). In this context, the combination of peer rejection and loneliness may result in increased involvement in resource-draining interactions typically associated with challenging peer relationships, such as managing conflicts. Consequently, these interactions could significantly deplete adolescents' self-control resources, leaving them with a heightened need to turn to the Internet as compensation.

However, when we incorporated the three indicators (i.e., popularity, peer rejection, and loneliness) and latent peer relationship profiles into the mediation model, the mediating effects associated with latent peer relationship profiles disappeared. Using latent profiles as independent variables in mediation analysis is a well-established research and statistical modeling practice, as evidenced by prior studies (e.g., Bakac et al., 2023; Boiché & Stephan, 2014; Pfattheicher et al., 2020). Nevertheless, it's important to note that in these earlier studies, the main effects of the individual indicators were not concurrently considered alongside the latent profiles. It's worth emphasizing that the latent profiles capture the interactions among the indicators of these profiles. If the person-centered approach is relevant, these latent profiles should explain additional variance, even after controlling for the main effects of indicators (Bouckennooghe et al., 2019). Therefore, findings from a person-centered approach should be interpreted with caution when not accounting for the main effects of indicators.

4.2. Implications

These findings have the potential to make meaningful contributions to the existing body of research in various ways. Firstly, from a methodological perspective, our results support the added value of a person-centered approach as it provides insights into distinct mechanisms within specific groups of individuals (von Eye & Bogat, 2006). By examining individual characteristics, such as peer relationships, researchers can identify unique patterns and mechanisms that may not be evident in traditional variable-centered approaches. Consequently, the combination of variable-centered and person-centered approaches can yield a more comprehensive understanding of the links between peer relationships (or other multidimensional constructs) and internet addiction, along with the underlying mechanisms at play. In future research, it is advisable also to consider the main effects of indicators when investigating the impact of latent profiles to assess their meaningfulness.

Secondly, these findings might contribute to the compensatory internet use theory by shedding light on the underlying compensatory mechanisms. Motivations for excessive internet use may not solely be rooted in changing environments (e.g., seeking more social ties or

escaping stressful lives; Kardefelt-Winther, 2014a) but also in changing oneself, such as enhancing self-evaluation and promoting self-control. Future research could explore additional mechanisms within the compensatory internet use theory.

Finally, the mechanisms uncovered in the current study have practical implications for interventions targeting adolescent internet addiction. It's worth emphasizing that while the Internet can offer opportunities for compensation and social support in the context of poor peer relationships (particularly peer rejection and loneliness), it cannot fully replace offline social relationships and face-to-face interactions. Debates persist regarding the impact of online interactions on the quantity and quality of communication among offline friends (Pouwels et al., 2021; Valkenburg & Peter, 2011) and the authenticity of online friendships (Amichai-Hamburger et al., 2013). Therefore, individuals must seek equilibrium between their online and offline lives and cultivate effective coping strategies to address the root causes of difficulties in their peer relationships. For example, implementing peer support programs with the goal of modification of social skills within schools or community settings (Lee et al., 2015) could be an effective strategy. Such programs would create opportunities for adolescents to connect with and support one another, fostering a sense of belonging and reducing feelings of social isolation.

4.3. Limitations

Several limitations to the present study should be noted. Firstly, our research was conducted solely in China and did not consider cultural aspects in our model, limiting our findings' generalizability. Notably, cultural factors play a significant role in shaping peer relationships (Chen et al., 2018). For example, in China, a collectivistic culture, self-control is highly valued to maintain interpersonal and group harmony. However, in Western societies, behavioral control may be less emphasized (Chen et al., 2018). Therefore, it is important to examine whether promoting self-control is a mechanism for coping with poor peer relationships through the Internet applies to other cultural contexts, particularly Western cultures.

Secondly, our study framed its findings within the framework of the compensation hypothesis. Notably, the popular peer status profile emerged as a protective factor against internet addiction, yielding a "rich-get-richer" effect that aligns with the enhancement hypothesis (Kraut et al., 2002). In line with this hypothesis, Zywicki and Danowski (2008) found that extroverted individuals on platforms like Facebook tend to attain popularity with comparatively less effort than their introverted counterparts. Therefore, in future research, it will be essential to delve deeper into the interplay between these profiles and the two hypotheses, as this holistic approach will contribute to a more comprehensive understanding of the nuanced motivations driving internet use during adolescence.

Thirdly, the use of cross-sectional data in this study presents inherent limitations, preventing the establishment of causal relationships and directionality of associations. To enhance robustness and depth, future research should employ longitudinal designs (e.g., longitudinal latent profile analysis) and experimental methods with diverse samples to seek to replicate the current study's findings and strengthen the evidence for causality.

Last but not least, it is important to acknowledge the potential influence of common method bias. All major variables in our study, including peer relationships, self-annoyance, self-control, and internet addiction, were self-reported. Future research should consider incorporating other sources of information, such as reports from peers, parents, or teachers, to offer a more holistic understanding of these constructs.

5. Conclusion

Using variable-centered and person-centered approaches, we

investigated the direct and indirect effects of peer relationships on internet addiction within the compensatory internet use framework. This study has important practical implications, as educators and practitioners can better identify adolescents with internet addiction and design targeted Internet interventions to address the underlying issues related to poor peer relationships.

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CRediT authorship contribution statement

Baobao Zhao: Conceptualization, Visualization, Formal analysis, Data curation. **Cancan Jin:** Funding acquisition, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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